

# Microsoft 365 SaaS Subscription & Churn Analytics

End-to-End Portfolio Project Case Study | Excel • SQL • Python • Power BI

\$445.8M

TOTAL PORTFOLIO ARR

5,000

CUSTOMER  
ACCOUNTS

7.80%

ANNUAL CHURN RATE

-65.2%

COPILOT CHURN  
REDUCTION

\$4.45M

ARR PRESERVED PER  
1% CHURN CUT

## 1. Microsoft Excel: Financial & What-If Modeling

- Dynamic KPI Model:** Built automated summary calculations using SUMIFS, COUNTIFS, and AVERAGEIFS.
- Interactive Customer Lookup Engine:** Created an account search dashboard utilizing XLOOKUP for instant churn risk inspection.
- What-If Sensitivity Analysis:** Modeled revenue preservation across 5 retention improvement scenarios (0.5% to 5.0%).

## 2. SQL / T-SQL: Data Warehousing & Analytics

- Star Schema Architecture:** Designed DDL with Dim\_Customers, Dim\_Plans, and Fact\_Subscriptions.
- Window Functions:** Computed 30-day moving revenue trends using AVG() OVER (ROWS BETWEEN 29 PRECEDING AND CURRENT ROW).
- Dense Ranking:** Identified top enterprise revenue accounts per industry without skipping ranks using DENSE\_RANK().
- Customer Lifetime Value:** Computed LTV by plan tier in SQL.

## 3. Python: Statistical Testing & Machine Learning

- Hypothesis & A/B Testing:** Validated that Copilot AI reduces churn from 10.38% to 3.61% ( $\chi^2 = 74.28$ ,  $p < 0.001$ ).
- Random Forest Classifier:** Trained churn risk model achieving **ROC-AUC = 0.7868**.
- Predictive Drivers:** Identified Teams adoption % (20.3%), Auto-Renewal flag (19.6%), and Support Tickets (13.5%) as top churn indicators.

## 4. Power BI & DAX: Business Intelligence

- Star Schema Data Modeling:** 1-to-Many single direction relationships optimized for Power BI VertiPaq memory.
- Advanced DAX Library:** Implemented Time Intelligence (SAMEPERIODLASTYEAR), filter overrides with CALCULATE(), and safe division with DIVIDE().
- Churn Early Warning Dashboard:** Visualized high-risk account matrices and cross-filtering telemetry.

## Interview Talking Points & Business Recommendations

| Finding / Observation    | Data Proof                                       | Recommended Business Action                                                |
|--------------------------|--------------------------------------------------|----------------------------------------------------------------------------|
| Copilot Drives Retention | Churn dropped from 10.38% → 3.61% ( -65% Churn ) | Offer 30-day free Copilot trial to at-risk accounts during renewal window. |

|                              |                                                                              |                                                                                     |
|------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Teams Usage Cliff</b>     | Accounts with <35% Teams usage have 78% higher churn likelihood              | Trigger automated customer success check-in when usage drops below 40%.             |
| <b>High Support Friction</b> | Accounts with $\geq 3$ tickets show elevated churn risk ( <b>High Risk</b> ) | Implement priority escalation routing for enterprise accounts with >2 open tickets. |